Course 02323 Introduction to Statistics & 02402 Statistics (Polytechnical Foundation)

Lecture 4: Confidence intervals/Interval Estimation

DTU Compute
Technical University of Denmark
2800 Lyngby – Denmark

Overview

1. Summary of Statistical Inference
2. Introduction to confidence intervals
3. Distribution of the sample mean
   - Standard error of the mean, SEM
   - The t-distribution
4. Confidence interval (CI) for μ
   - Example: Heights
5. Confidence interval for variance and standard deviation
6. The Central Limit Theorem (CLT)
7. Examples

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The formal framework for *statistical inference*

From Chapter 1 of the book:

- An *observational unit* is the single entity/level about which information is sought (e.g. a person) (**Observation Unit**)
- The *statistical population* consists of all possible "measurements" on each *observational unit* (**Population**)
- The *sample* from a statistical population is the actual set of data collected. (**Sample**)

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Terminology and concepts:

- μ and σ are *parameters* that describe the population
- $\hat{\mu} = \bar{x}$ is the *estimate* for μ (specific outcome value)
- $\bar{X}$ is the *estimator* for μ (now seen as a random variable)
- The concept of *statistic* is a common term for both ($\bar{x}$ and $\bar{X}$)

The formal framework for *statistical inference* - Example

From Chapter 1 of the book: Modified height example

We measure the height of 10 random individuals in Denmark.

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The sample:

The 10 observations: $x_1, \ldots, x_{10}$.

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The population:

The heights of all people in Denmark.

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We measure the height of 10 random individuals in Denmark.

The sample:

The 10 observations: $x_1, \ldots, x_{10}$.

The population:

The heights of all people in Denmark.

The observational unit:

One individual.

Statistical inference: Learning from data

Learning from data:

We want to deduce the parameter values for the underlying population.

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Important in this regard:

The sample must meaningfully be *representative* of a well-defined population.

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Learning from data:

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Important in this regard:

The sample must meaningfully be *representative* of a well-defined population.

How do you ensure this?

For example, by ensuring that the sample is fully *randomly selected*.

Random sampling

Definition 3.12:

- A random sample from an (infinite) population: The random variables $X_1, X_2, ..., X_n$ constitute a random sample of size n from the infinite population if:

  1. All the random variables have the same distribution
  2. The n random variables are independent

Random sampling

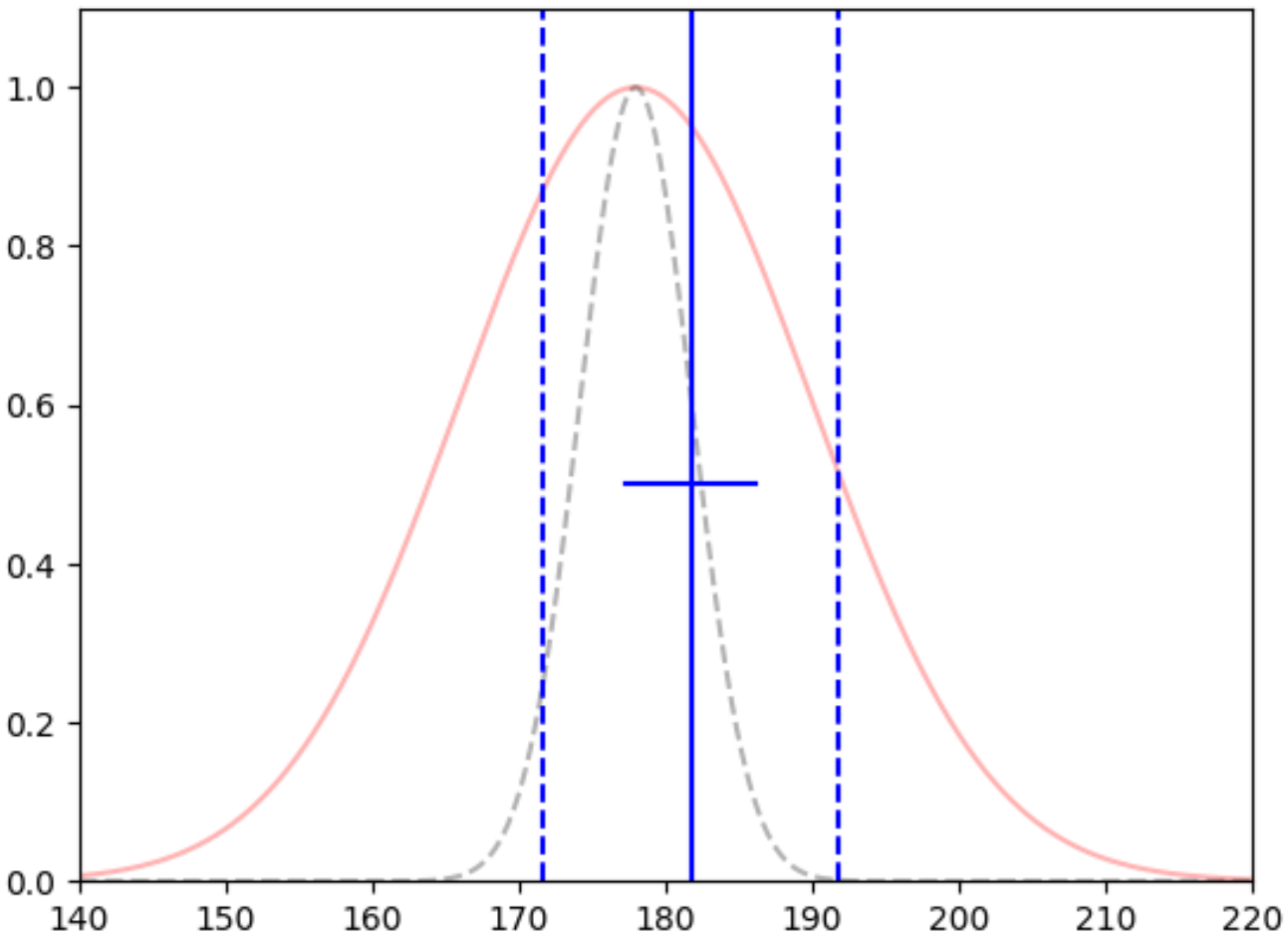
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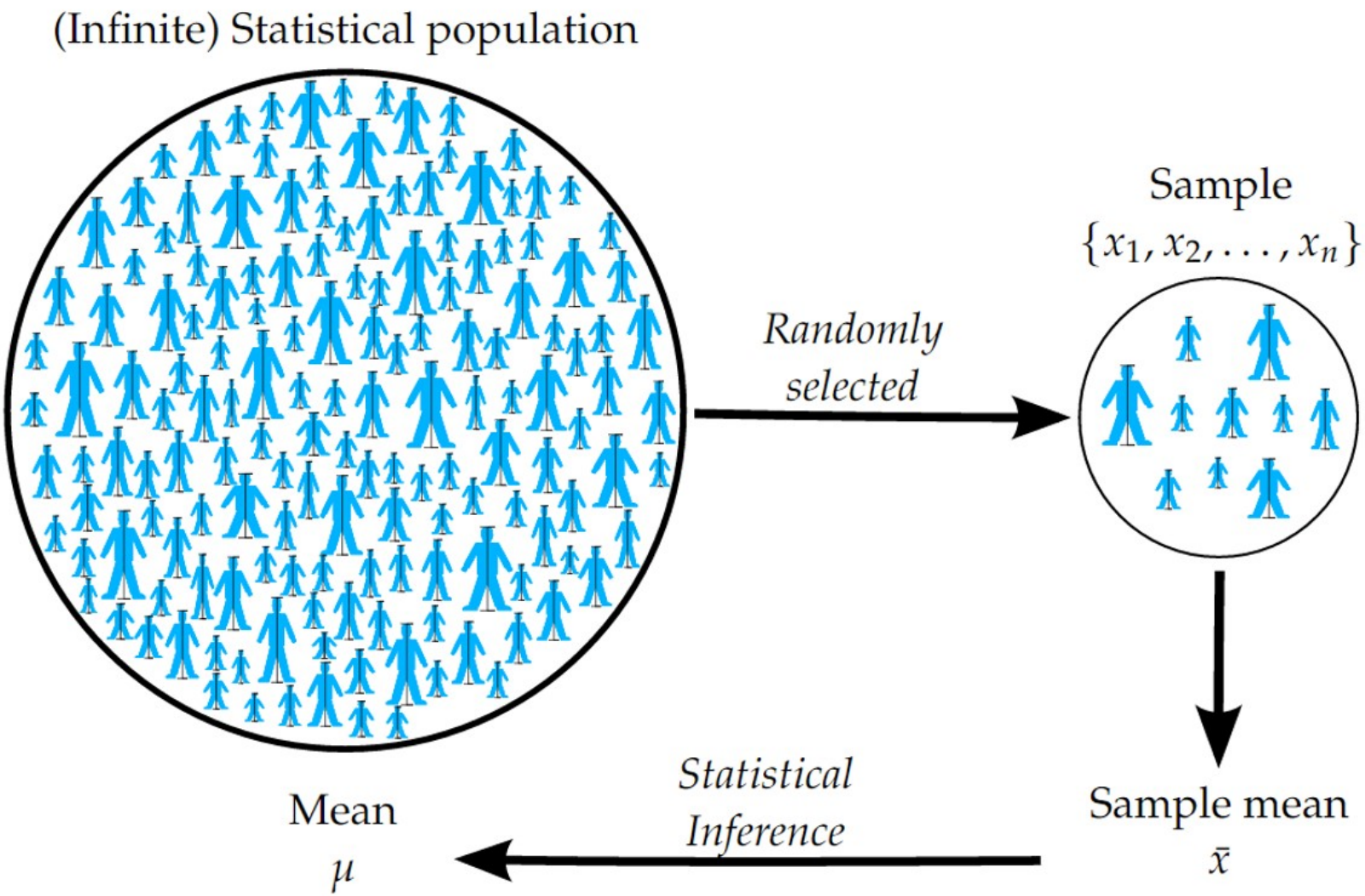
What does this mean?

1. All observations must come from the same population
2. They must NOT share information with each other (e.g. if you had sampled entire families instead of individuals)

Statistical inference: Learning from data



The population and the sample



What does it mean?

Statistical inference: Learning from data

- We *assume* a *model* for the population, e.g.: $X \sim N(\mu, \sigma^2)$
- We *assume* that our observations (data) constitute a *random sample* (a representative sample).
- The model has some *parameters*, e.g.: μ and σ
- We now want to *estimate* the parameters based on the data in the sample
  - $\hat{\mu} = \bar{x}$ is the *estimate* for μ (specific outcome value)
  - $\bar{X}$ is the *estimator* for μ (now seen as a stochastic variable)
  - $\hat{\sigma}^2 = s^2$ is the *estimate* for σ^2 (specific outcome value)
  - S^2 is the *estimator* for σ^2 (now seen as a stochastic variable)
- Based on our assumptions, we can theoretically predict how our estimates will behave – from here we work backwards to make statements about the population. Conclusions from statistical inference thus rely on the assumptions about the model holding true.

What is the uncertainty?

How much can we trust estimates?

Can we quantify how good the estimates are?

Today: *Standard error* and *confidence intervals*

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Example - Height of 10 students:

Sample, $n = 10$:

168 161 167 179 184 166 198 187 191 179

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Sample mean and standard deviation:

$$\bar{x} = 178$$

$$s = 12.21$$

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$$\bar{x} = 178$$
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Estimates for the population mean and standard deviation:

$$\hat{\mu} = 178$$
$$\hat{\sigma} = 12.21$$

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But how reliable are these estimates?

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Estimates for the population mean and standard deviation:

$$\hat{\mu} = 178$$
$$\hat{\sigma} = 12.21$$

But how reliable are these estimates?

NEW: **Confidence Intervals:**

$$\hat{\mu} = \ 178 \ \ [169.3;\ 186.7]$$
$$\hat{\sigma} = 12.21 \ \ [8.4;\ 22.3]$$

Interval Estimation: Confidence Intervals

- The purpose of an interval estimate is to provide information about how close the point estimate, provided by the sample, is to the value of the population parameter.
- The general form of an interval estimate of a population mean is

$$\bar{x} \pm \text{Margin of error}$$

Python: (Empirical) Distribution of the sample mean

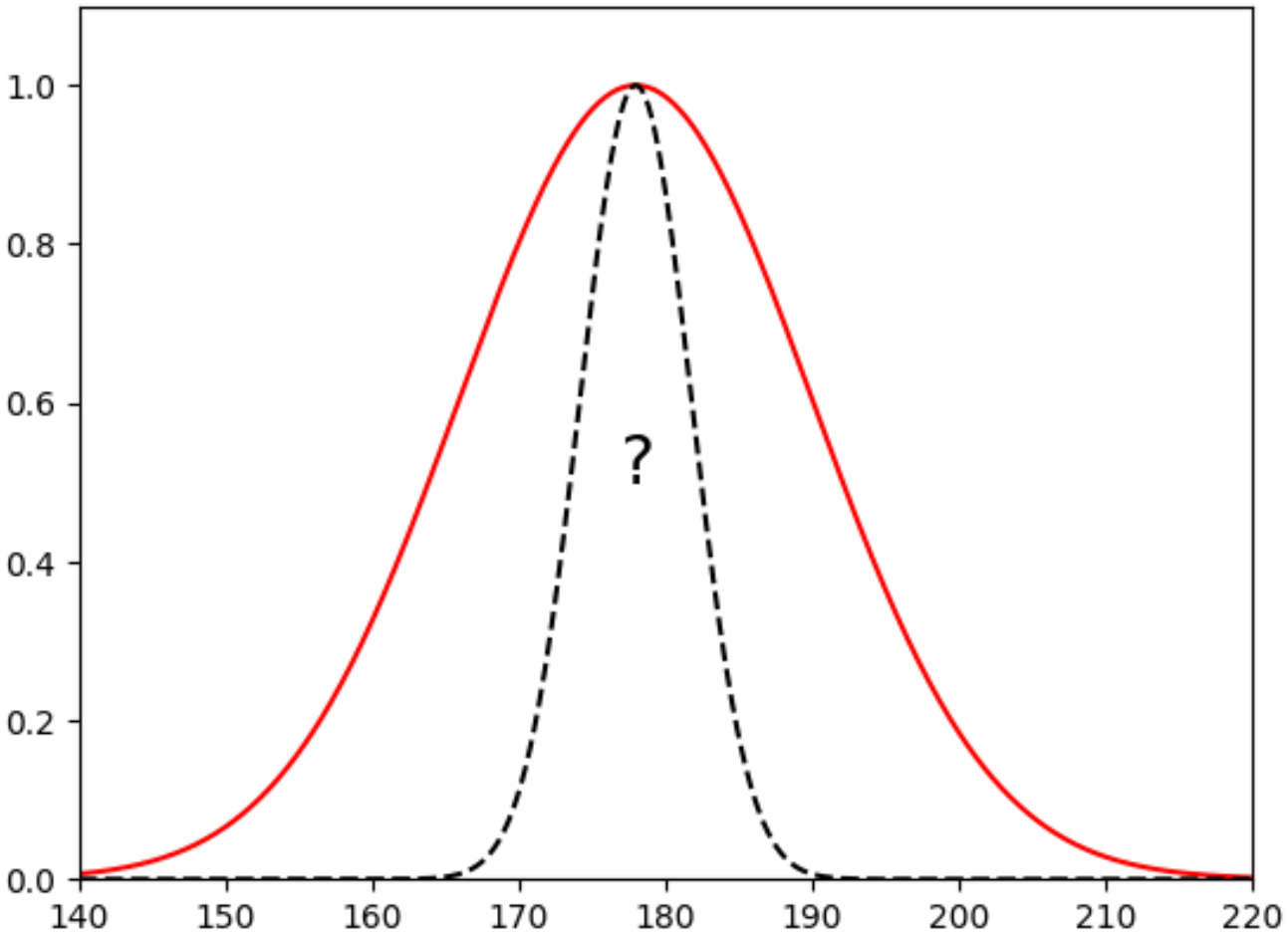
- Go to today's Python notebook in VS Code
  - "Simulation: Distribution of the sample mean"



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(Theoretical) distribution of sample mean



What is the theoretical distribution of the sample mean?

Theorem 3.3: Distribution of the sample mean of i.i.d. normal random variables

Now the sample mean, $\bar{X} = \frac{1}{n}(X_i + ... + X_n)$, itself is a **random variable** (with its own distribution).

The distribution of $\bar{X}$

Assume that $X_1, \ldots, X_n$ are independent and identically distributed (*i.i.d.*) normal random variables, $X_i \sim N(\mu, \sigma^2), i = 1, \ldots, n$, then:

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i \quad \textit{and} \quad N\left(\mu, \frac{\sigma^2}{n}\right)$$

The mean and variance follow from the rules of calculation

The mean of $\bar{X}$:

$$\mathsf{E}[\bar{X}] = \mathsf{E}[\frac{1}{n}X_i + \ldots + \frac{1}{n}X_n] = \frac{1}{n}\sum_{i=1}^{n}\mathsf{E}[X_i] = \frac{1}{n}\sum_{i=1}^{n}\mu = \frac{1}{n}n\mu = \mu$$

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The mean of $\bar{X}$:

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The variance of $\bar{X}$:

$$\mathsf{V}[\bar{X}] = \mathsf{V}[\frac{1}{n}X_i + \ldots + \frac{1}{n}X_n] = \frac{1}{n^2}\sum_{i=1}^{n}\mathsf{V}[X_i] = \frac{1}{n^2}\sum_{i=1}^{n}\sigma^2 = \frac{1}{n^2}n\sigma^2 = \frac{\sigma^2}{n}$$

The mean and variance follow from the rules of calculation

The mean of $\bar{X}$:

$$\mathsf{E}[\bar{X}] = \mathsf{E}[\frac{1}{n}X_i + \ldots + \frac{1}{n}X_n] = \frac{1}{n}\sum_{i=1}^{n}\mathsf{E}[X_i] = \frac{1}{n}\sum_{i=1}^{n}\mu = \frac{1}{n}n\mu = \mu$$

The variance of $\bar{X}$:

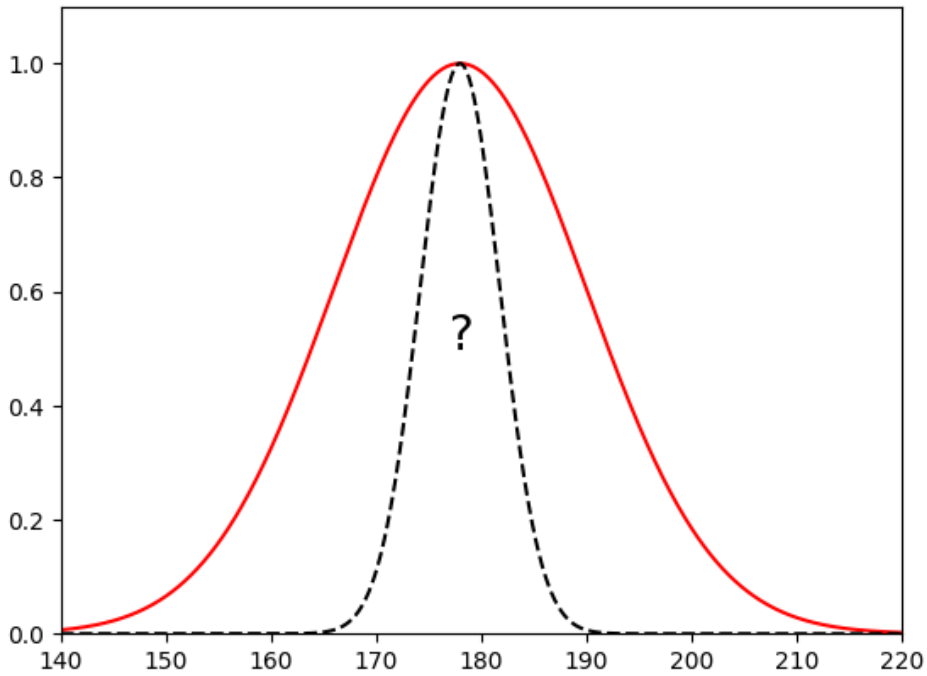
$$\mathsf{V}[\bar{X}] = \mathsf{V}[\frac{1}{n}X_i + \ldots + \frac{1}{n}X_n] = \frac{1}{n^2}\sum_{i=1}^{n}\mathsf{V}[X_i] = \frac{1}{n^2}\sum_{i=1}^{n}\sigma^2 = \frac{1}{n^2}n\sigma^2 = \frac{\sigma^2}{n}$$

Normality of $\bar{X}$:

A linear combination of normally distributed random variables will also be normally distributed (with mean and variance as given above).

(Theoretical) distribution for the sample mean

If data follow the red normal distribution, the sample mean will follow the black – narrower – normal distribution.

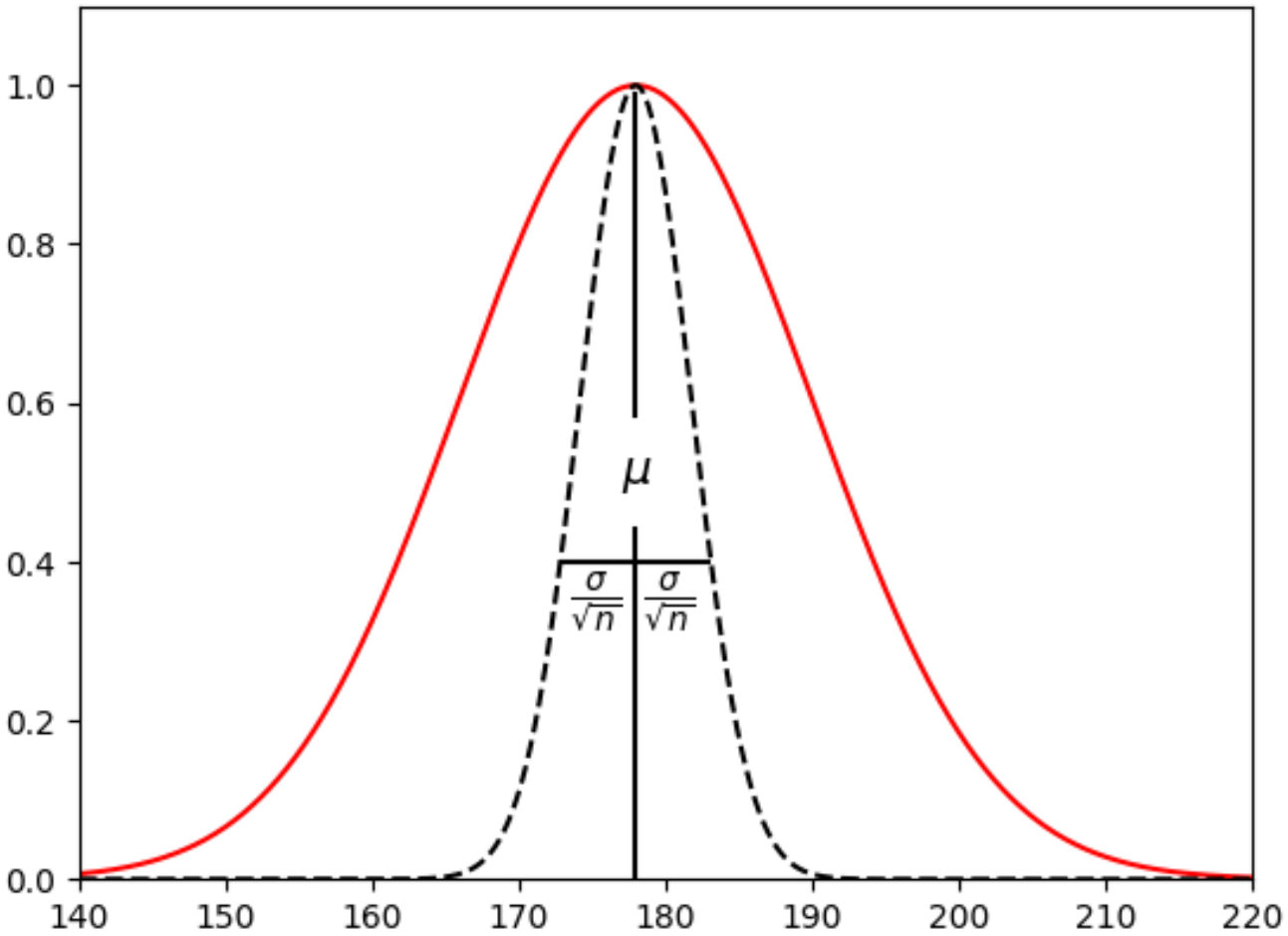


Kahoot!

What are typical values of $\bar{X}$ in the example above?

(x2)

(Theoretical) distribution for the mean



Distribution of sample mean

How large is the error $(\bar{X} - \mu)$?

When we estimate μ with $\bar{X}$, we must expect some uncertainty – or "error".
We can describe the error as a **stochastic variable**: $error = (\bar{X} - \mu)$.

- The average error is zero: $\mathrm{E}[error] = \mathrm{E}[\bar{X} - \mu] = \mu - \mu = 0$
- But the variance of the error is: $\mathrm{V}[error] = \mathrm{V}[\bar{X} - \mu] = \mathrm{V}[\bar{X}] = \frac{\sigma^2}{n}$
- The error is normally distributed: $error \sim N(0, \sigma^2/n)$
- And thus the **standard deviation of the error** is: $\frac{\sigma}{\sqrt{n}}$

Kahoot!
(x2)

Standard Error of the Mean

In practice, σ is not known, but $\frac{\sigma}{\sqrt{n}}$ is estimated with: $\frac{s}{\sqrt{n}}$. The latter has its own name, namely "Standard error of the mean":

Definition 3.7 Standard Error of the mean

Given a sample $X_1, \ldots, X_n$, the *Standard Error of the Mean* is defined as

$$\sigma_{\bar{x}} = \frac{S}{\sqrt{n}}. \tag{3-9}$$

It can also be read as the *Sampling Error* of the mean, and can be called the standard deviation of the *sampling distribution* of the mean.

Kahoot!

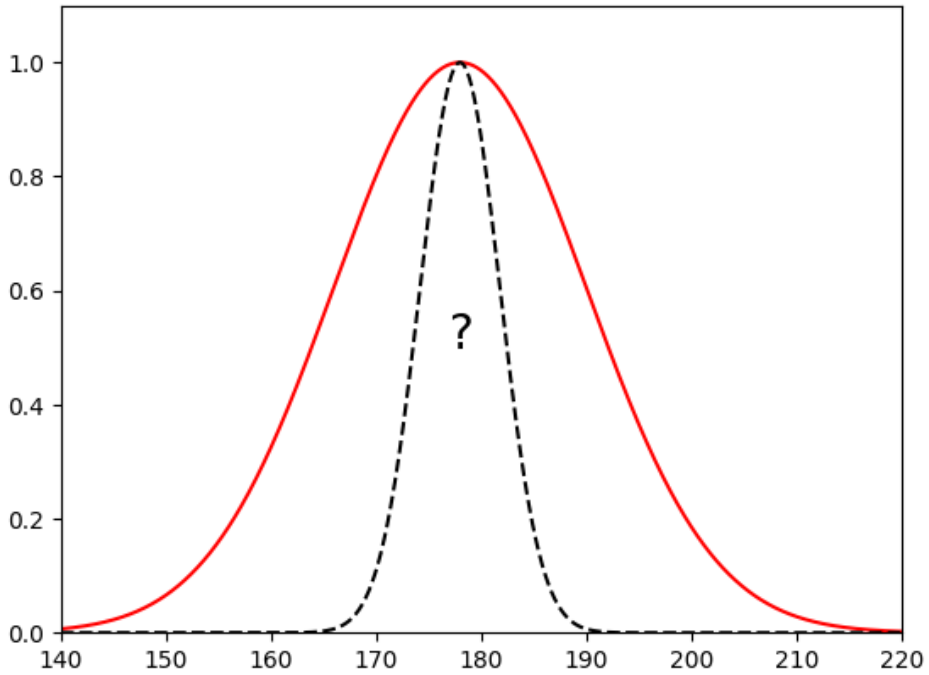
(x2)

What can we expect about the value of $\bar{X}$?

In practice, we only have one sample and therefore only one observation of $\bar{X}$.

But based on our theory, we *know* that $\bar{X}$ *with 95% probability* will lie in the interval:

$$[\mu - 1.96\frac{\sigma}{\sqrt{n}}; \mu + 1.96\frac{\sigma}{\sqrt{n}}]$$



Standardized version of the same things

Theorem 3.4 The distribution of the σ-standardized mean of normal random variables

Assume that $X_1, \ldots, X_n$ are independent and identically normally distributed random variables, $X_i \sim N\left(\mu, \sigma^2\right)$ where $i = 1, \ldots, n$, then

$$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} \sim N\left(0, 1^2\right). \tag{3-7}$$

That is, the standardized sample mean Z follows a standard normal distribution.

The probability that Z lies in the interval $[-1.96; 1.96]$ is 95%.

These numbers (-1.96 and $+1.96$) are quantiles in the standard normal distribution:

$P(Z \leq -1.96) = 0.025$

$P(Z \leq +1.96) = 0.925$

Kahoot!

(x2)

What are "extreme values" of $|Z|$?

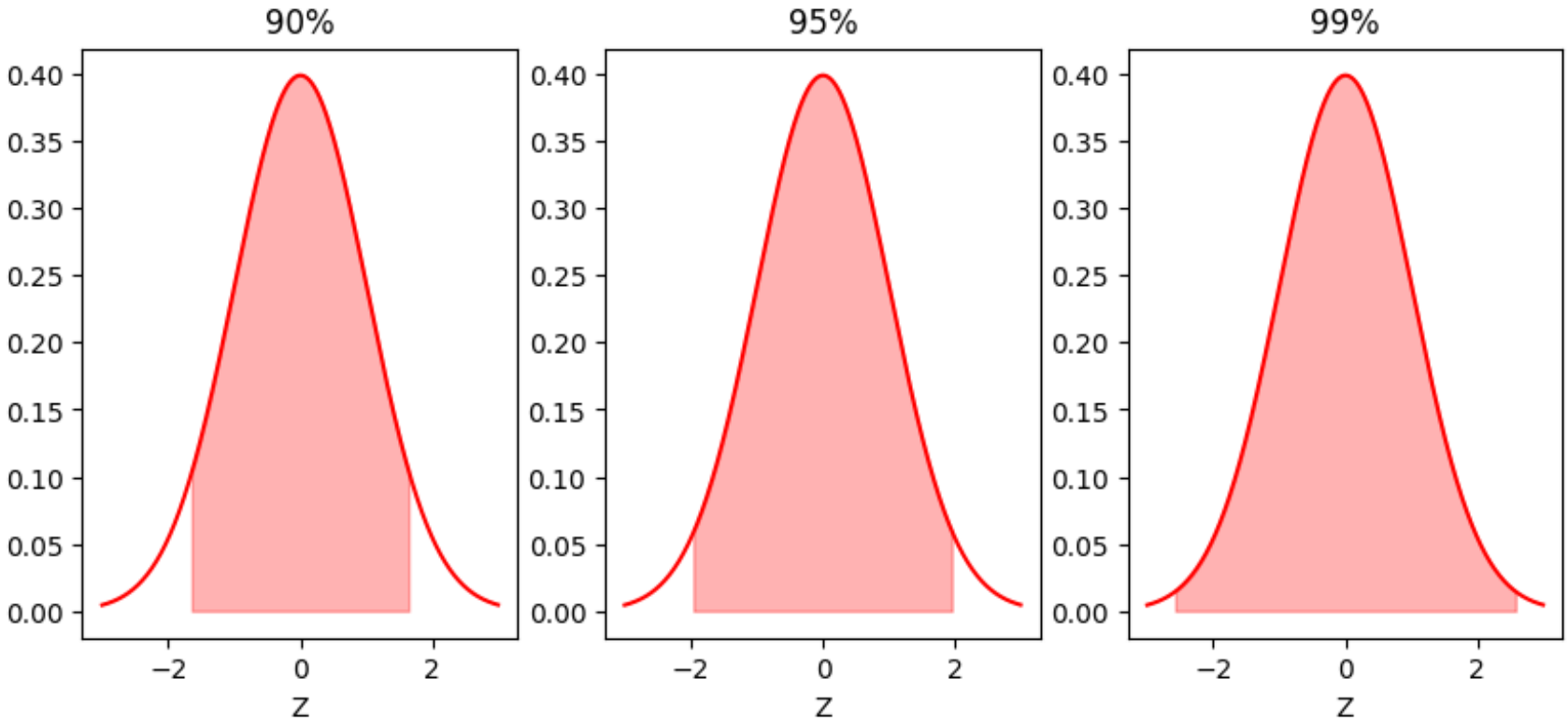
$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N\left(0, 1^2\right)$$

Large values of $|Z|$ (also called "**extreme values**") correspond to situations where $\bar{X}$ is very far from μ – i.e. one has been "unlucky" with the sample in the sense that the estimate ($\hat{\mu} = \bar{x}$) lies far from the true value μ.

In the following, we will build on an assumption that one has probably not been "too unlucky" with the specific sample. Exactly how unlucky(/lucky) is quantified with a "**significance level**".

Z-intervals

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N\left(0, 1^2\right)$$



The "extreme" cases (we could also say "extremely unlucky" cases) lie outside the red interval.

Confidence interval (CI) for μ: σ assumed known

Use the correct z-distribution to construct the confidence interval:

For a sample $x_1, \ldots, x_n$ the $100(1-\alpha)\%$ confidence interval is given by:

$$\bar{x} \pm z_{1-\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

where $z_{1-\alpha/2}$ is the $100(1-\alpha/2)\%$ quantile from the z distribution or standard normal distribution.

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For a sample $x_1, \ldots, x_n$ the $100(1-\alpha)\%$ confidence interval is given by:

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where $z_{1-\alpha/2}$ is the $100(1-\alpha/2)\%$ quantile from the z distribution or standard normal distribution.

Most commonly using $\alpha = 0.05$:

The most commonly used is the 95% confidence interval:

$$\bar{x} \pm z_{0.975} \cdot \frac{\sigma}{\sqrt{n}}$$

Most commonly used confidence levels

Table: Values of $Z_{\alpha/2}$ for the most commonly used confidence levels

Confidence Level	α	$\alpha/2$	$Z_{\alpha/2}$
90%	0.10	0.05	1.645
95%	0.05	0.025	1.960
99%	0.01	0.005	2.576

Interval estimate of a population mean: σ unknown

We've estimated $\bar{x}$ and s^2 from our sample data.
We do not know the "true" μ and σ^2.
What do we do then?:
Apply t-distribution ($n-1$ degrees of freedom)
— the theory deals with uncertainty linked to s.

Theorem 3.5, The t-distribution

Instead, we consider the random variable: $T = \frac{\bar{X}-\mu}{S/\sqrt{n}}$

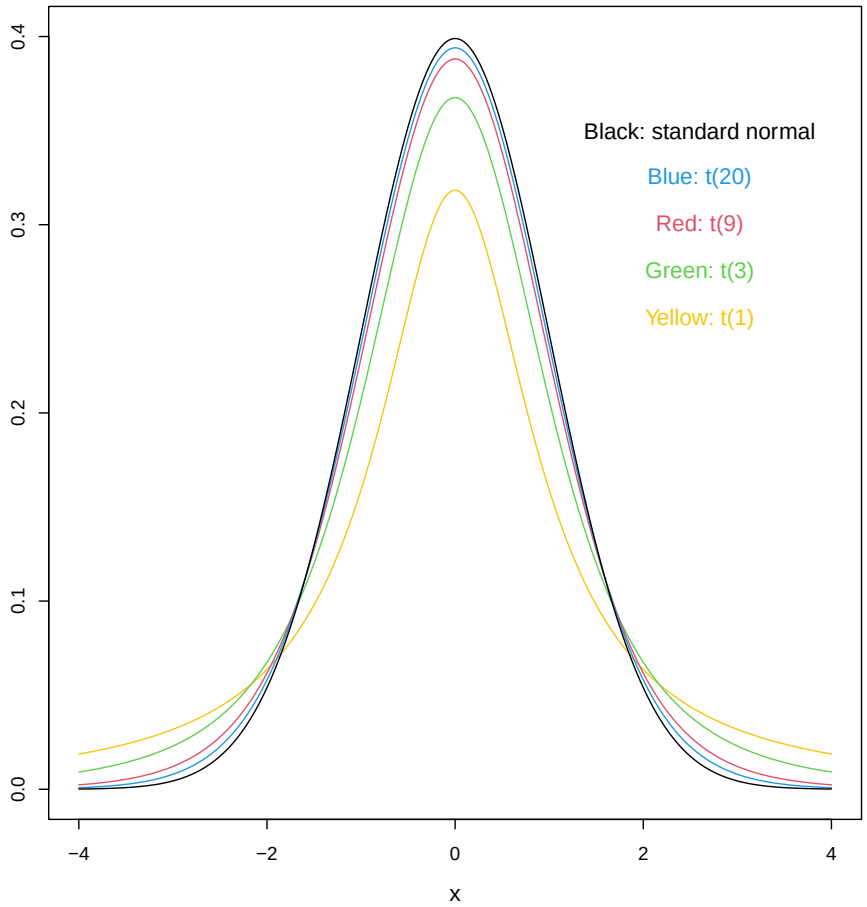
The t-distribution takes the uncertainty of s into account:

Assume that $X_1, \ldots, X_n$ are i.i.d. normal distributed random variables, where $X_i \sim N\left(\mu, \sigma^2\right)$ for $i = 1, \ldots, n$, then:

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t(n-1)$$

where $t(n-1)$ is the t-distribution with $n-1$ degrees of freedom.

Different t-distributions:



With the increase in sample size or df, (1) the shape of the distribution becomes narrower (2) Tails Become Thinner (3) t-distribution approaches Normal distribution (4) Critical t-values approach Z-values, (5) Confidence intervals become narrower

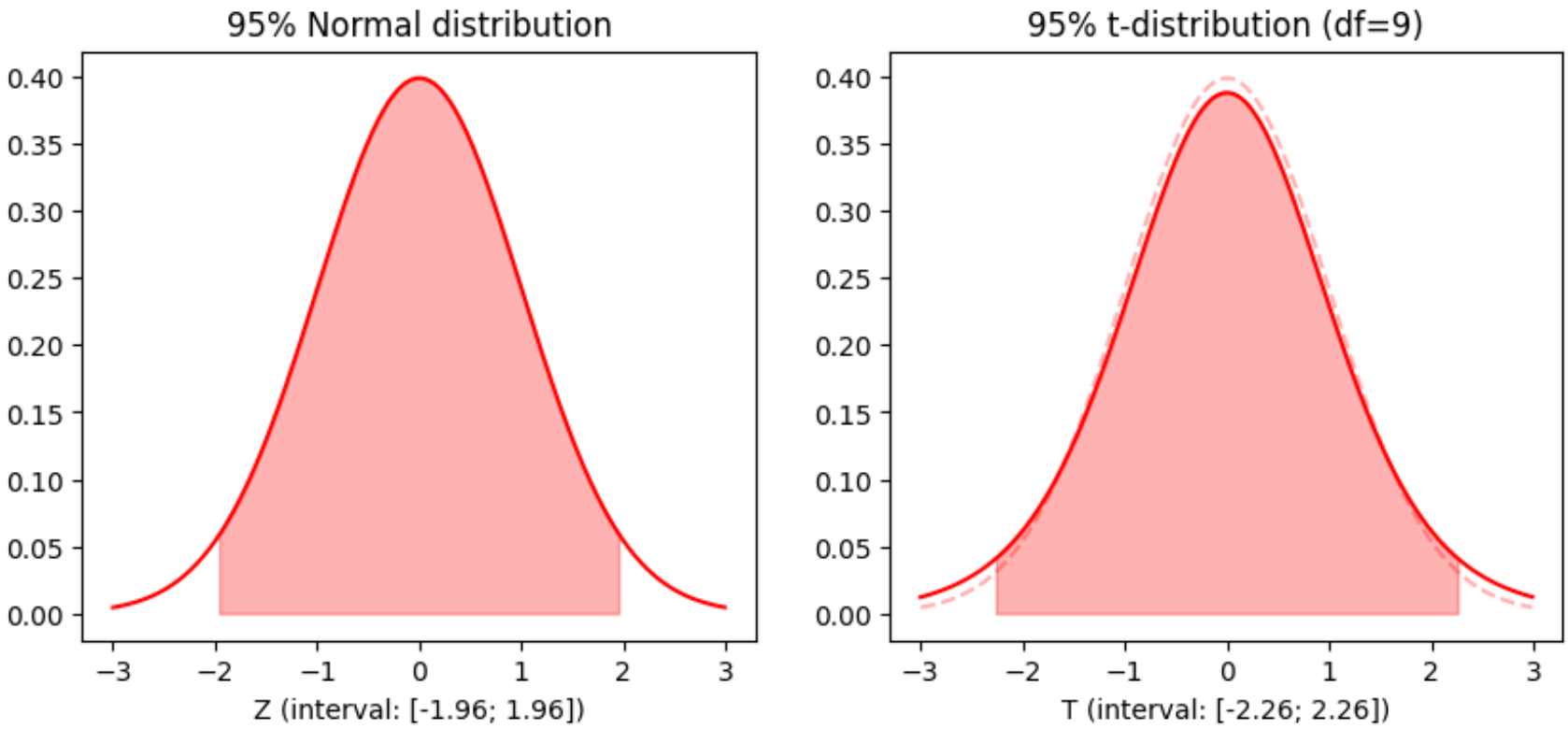
Python: t-distribution

`https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.t.html`

- Go to today's Python notebook in VS Code
  - "t-distribution"



t-distribution with 9 degrees of freedom and the standard normal distribution:



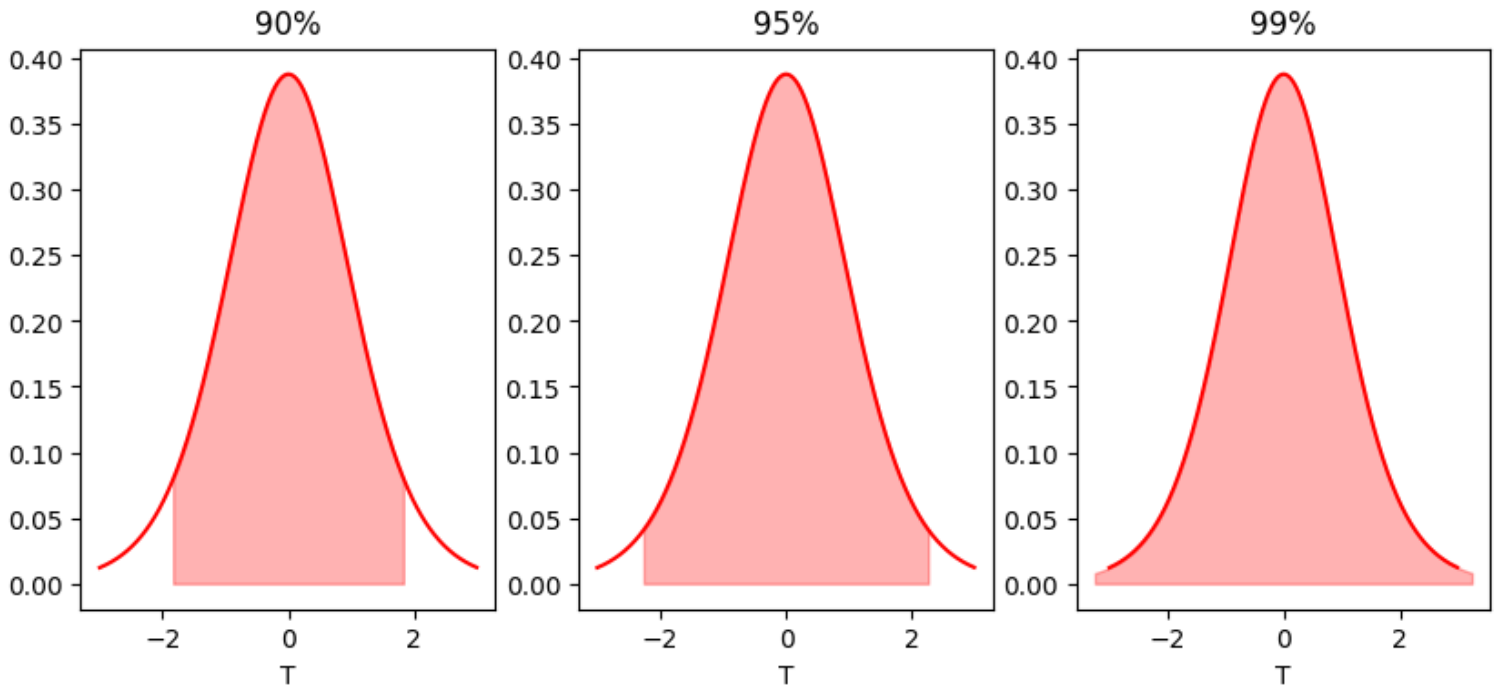
The t-distribution resembles the Normal distribution.

In the t-distribution the probability mass is spread a bit wider.

The width of the t-distribution depends on the number of degrees of freedom (df).

T-intervals

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t\,(n-1)$$



Large values of $|T|$ ("extreme values") correspond to situations where one has been "unlucky" with the sample, in the sense that the estimate $\hat{\mu} = \bar{x}$ lies far from the true value μ (possibly at the same time as s is very small).

Interval for μ

Again, a ***"significance level"*** α is chosen, and we *work backwards* to obtain an interval for μ.

Example with interval covering 95%:

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} > t_{0.025} \quad \text{and} \quad T = \frac{\bar{X} - \mu}{S/\sqrt{n}} < t_{0.975}$$

Convert the two inequalities into a corresponding interval for μ:

(Note that $t_{0.025} = -t_{0.975}$)

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Confidence interval for μ

Method 3.9 The one sample confidence interval for μ

For a sample $x_1, \ldots, x_n$ the $100(1-\alpha)\%$ confidence interval is given by

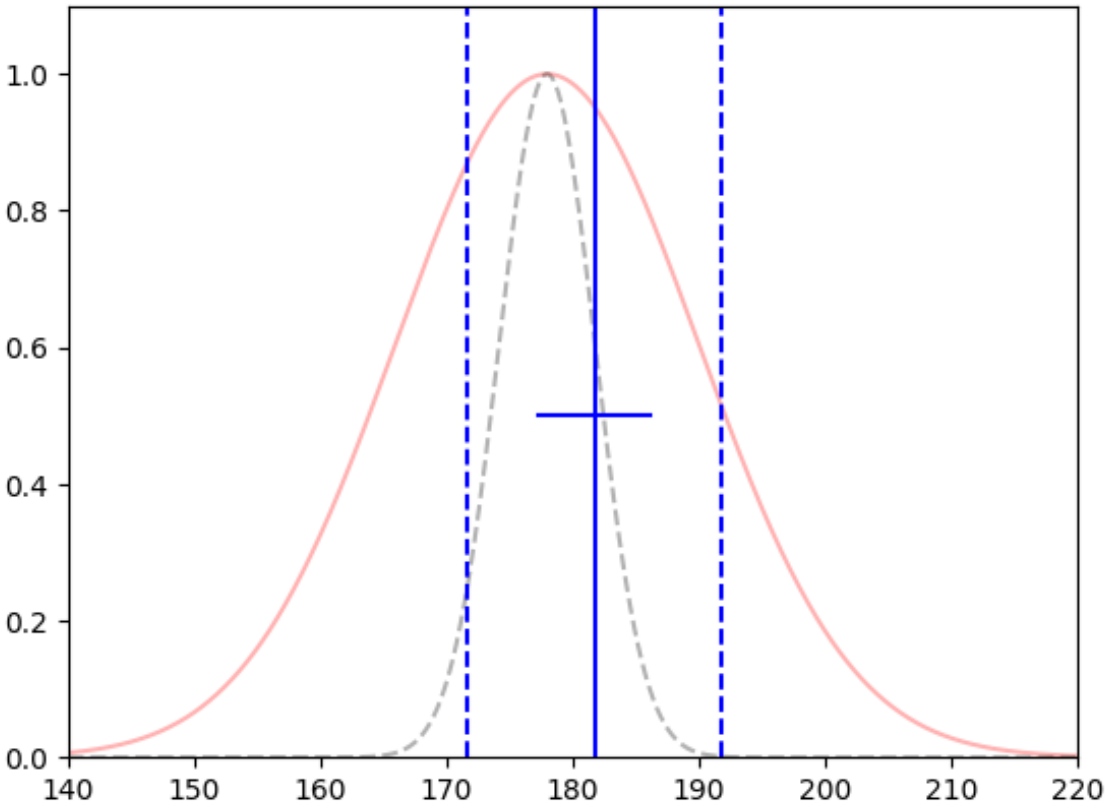
$$\bar{x} \pm t_{1-\alpha/2} \cdot \frac{s}{\sqrt{n}}, \tag{3-10}$$

where $t_{1-\alpha/2}$ is the $(1-\alpha/2)$ quantile from the t-distribution with $n-1$ degrees of freedom.[a]
Most commonly used is the 95%-confidence interval:

$$\bar{x} \pm t_{0.975} \cdot \frac{s}{\sqrt{n}}. \tag{3-11}$$

Visualization of the single case

We have now estimated $\bar{x}$ and s^2 from our sample data.
We do not know the "true" μ and σ^2, but we can state a **confidence interval** for μ.
Together with a confidence interval one should always state the significance level α.



Kahoot!
(x3)

'Repeated sampling' interpretation

If $\alpha = 0.05$:

$$P\left(\bar{X} - t_{0.975}\frac{S}{\sqrt{n}} < \mu < \bar{X} + t_{0.975}\frac{S}{\sqrt{n}}\right) = 0.95.$$

In the long run, we capture the true value in 95% of cases:

The confidence interval will vary in both width (s) and position ($\bar{x}$) if you repeat the study.

Kahoot!

(x4)

The significance level: α

Significance level "α":

A 95% confidence interval corresponds to $\alpha = 0.05$.

A 99% confidence interval corresponds to $\alpha = 0.01$, etc.

Often $\alpha = 0.05$ is chosen.

Quantiles in the t-distribution must be found with Python

When we need to calculate confidence intervals, we need quantiles from the t-distribution.

To find these, we need to know α (for a 95% confidence interval $\alpha = 0.05$) and we need to know the number of degrees of freedom $\nu = n - 1$ (where n is the sample size).

We must then find the $(1 - \alpha/2)$-quantile in a $t(n-1)$ distribution (this quantile is called "$t_{1-\alpha/2}$").

Example with $t_{0.975}$ (i.e. $\alpha = 0.05$) and $n = 30$ (a sample of 30 observations):

```
stats.t.ppf(q=0.975, df=29)
```

See also Appendix A.2.1 (section on t-distribution) in the book.

Height Example

- Go to today's Python notebook in VS Code
  - Example: find correct quantiles in t(9)-distribution
  - 95% interval estimate for students' height sample data
  - Visualize the confidence interval



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Python: (Empirical) Distribution of the sample variance

We have talked a lot about the sample **mean**, but what about the sample **variance**?

- Go to today's Python notebook in VS Code
  - "Simulation: Distribution of the sample variance"



- Note
  - The variance is always positive (squared unit) and does not follow a normal distribution (variance is a measure, not a random variable).
  - The distribution of variance is not symmetrical.

The distribution of the sample variance

We now consider the random variable: $\chi^2 = \frac{(n-1)S^2}{\sigma^2}$

In the expression for χ^2, the random variable S^2 is included

Theorem 2.81

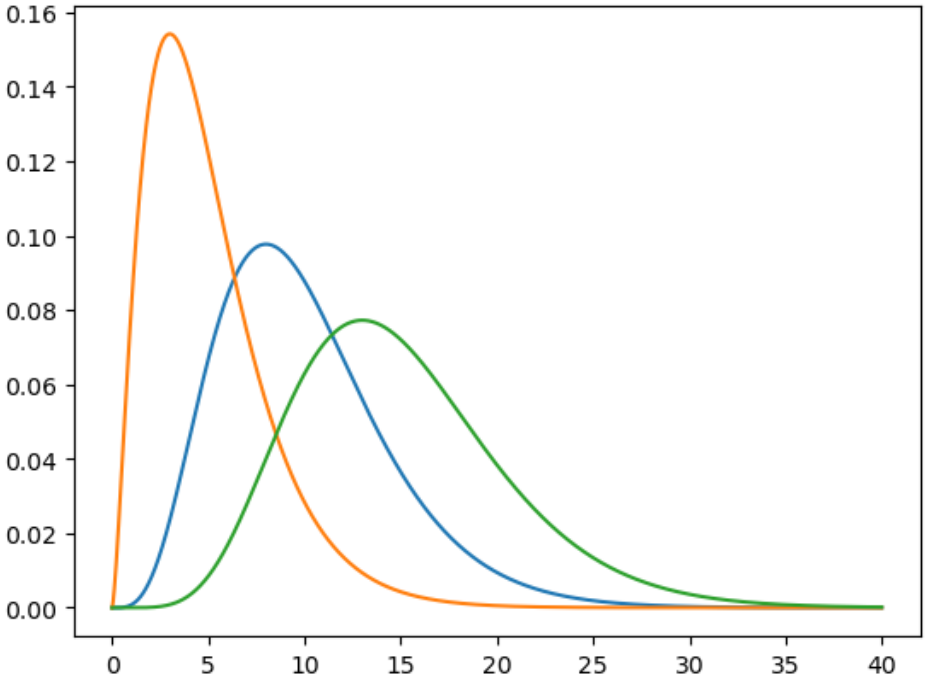
Given a sample of size n from the normal distributed random variables X_i with variance σ^2, then the sample variance S^2 (viewed as random variable) can be transformed into

$$\chi^2 = \frac{(n-1)S^2}{\sigma^2}, \tag{2-95}$$

which follows the χ^2-distribution with degrees of freedom $\nu = n - 1$.

Note the notation: The random variable "χ^2" follows a χ^2 distribution.

χ^2 distribution



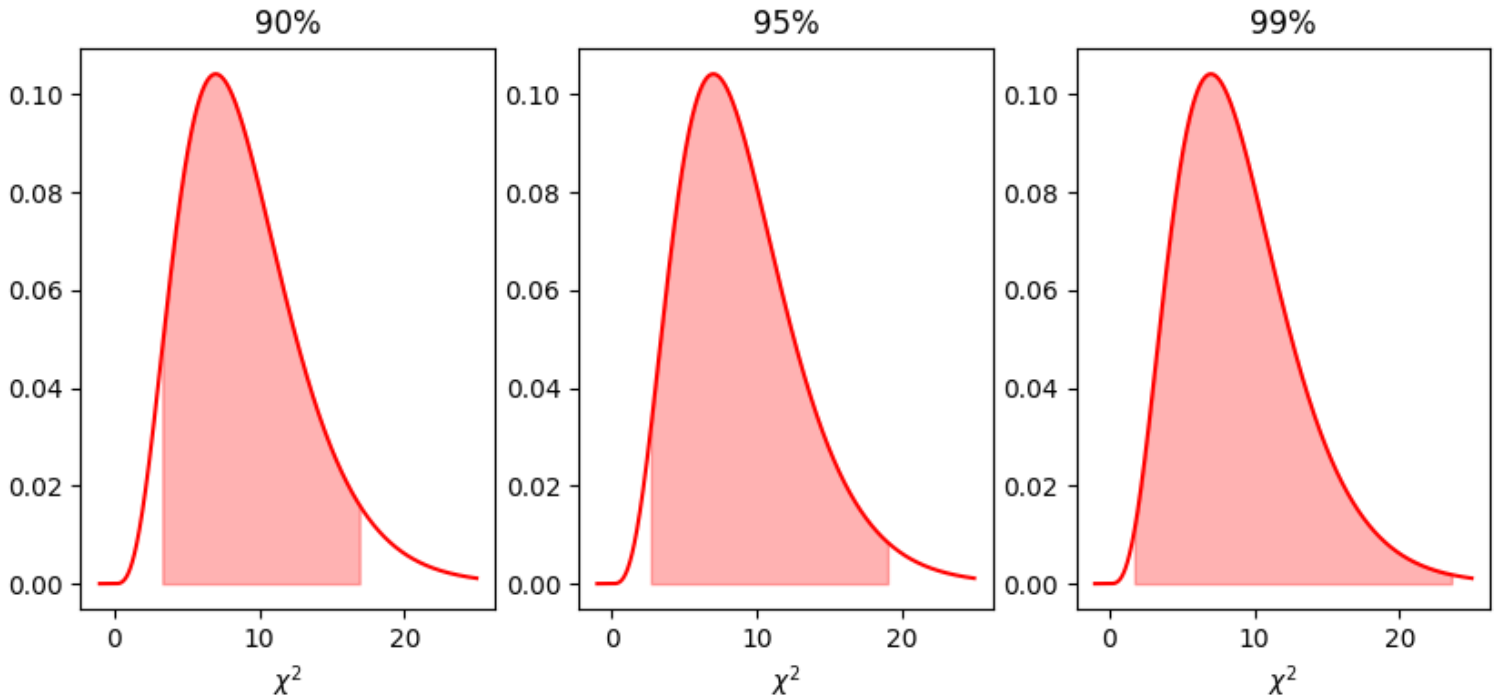
$$\chi^2 = \frac{(n-1)S^2}{\sigma^2}$$

The χ^2 distribution is not symmetric.

Can χ^2 become negative? When is χ^2 large/small? When is $\chi^2 = n-1$? What is $E[\chi^2]$?

χ^2 distribution

$$\chi^2 = \frac{(n-1)S^2}{\sigma^2}$$



Again: We need to choose a ***"significance level"*** (α).
Then, We can calculate interval bounds for σ^2 from the interval bounds on χ^2 (on the board).

Method 3.19: Confidence intervals for variance and standard deviation

Let $X_i \sim N(\mu, \sigma^2)$ for $i = 1, \ldots, n$ be independent (and identically distributed).

Variance:

A $100(1-\alpha)\%$ confidence interval for the variance σ^2 is given by:

$$\left[\frac{(n-1)s^2}{\chi^2_{1-\alpha/2}}; \frac{(n-1)s^2}{\chi^2_{\alpha/2}} \right],$$

where the quantiles come from a χ^2 distribution with $n-1$ degrees of freedom.

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Standard deviation:

A $100(1-\alpha)\%$ confidence interval for the standard deviation σ is:

$$\left[\sqrt{\frac{(n-1)s^2}{\chi^2_{1-\alpha/2}}}; \sqrt{\frac{(n-1)s^2}{\chi^2_{\alpha/2}}} \right].$$

Quantiles in the χ^2 distribution must be found with Python

When we need to calculate confidence intervals for σ^2, we need quantiles from the χ^2 distribution.

To find these, we must know α and the number of degrees of freedom $\nu = n - 1$ (n is the sample size).

The $(1-\alpha/2)$ quantile in a $\chi^2(n-1)$ distribution is called "$\chi^2_{1-\alpha/2}$".
The $(\alpha/2)$ quantile in a $\chi^2(n-1)$ distribution is called "$\chi^2_{\alpha/2}$".

Example with $\chi^2_{0.975}$ and $\chi^2_{0.025}$ (i.e. $\alpha = 0.05$) and $n = 30$ (a sample of 30 observations):

```
stats.chi2.ppf(q=0.975, df=29)
stats.chi2.ppf(q=0.025, df=29)
```

Kahoot!

(x2)

See also Appendix A.2.1 (section on the χ^2 distribution) in the book.

Python: Confidence interval for the variance estimate

Kahoot!

- Go to today's Python notebook in VS Code
  - "Example: Variance of student heights"

Notice that the confidence interval for the variance and standard deviation are not symmetric.



Example with confidence interval for s

A pastry chef needs a large number of uniform strawberries for a cake. He wants to buy strawberries from the supplier who can provide the most uniform berries – i.e., he wants the variation in the strawberries' diameter to be as small as possible.

As a test, he buys a tray with 23 strawberries and measures their diameter at the widest point. The strawberries' diameter has an average of 3 cm and a standard deviation of 0.5 cm.

The pastry chef now wants to calculate a 95% confidence interval for the standard deviation.

It is given that:

```
stats.chi2.ppf(0.025, df=22) = 3.31396
stats.chi2.ppf(0.975, df=22) = 6.06471
```

What is the significance level?

Calculate the confidence interval for s. Are the given quantiles the correct ones?

Overview

1. Summary of Statistical Inference
2. Introduction to confidence intervals
3. Distribution of the sample mean
   - Standard error of the mean, SEM
   - The t-distribution
4. Confidence interval (CI) for μ
   - Example: Heights
5. Confidence interval for variance and standard deviation
6. The Central Limit Theorem (CLT)
7. Examples

Theorem 3.14: The Central Limit Theorem (CLT)

The distribution of the sample mean approaches a normal distribution as the sample size n becomes sufficiently large, regardless of the shape of the population distribution.

Let $\bar{X}$ be the average of a randomly drawn sample of size n taken from a population with mean μ and variance σ^2. Then, the distribution of

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

approaches the standard normal distribution, $N(0, 1^2)$, as $n \to \infty$.

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$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

approaches the standard normal distribution, $N(0, 1^2)$, as $n \to \infty$.

That is, if n is large enough, we can (approximately) assume:

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1^2).$$

English: *Central Limit Theorem* (CLT)

Python: CLT in action

- Go to today's Python notebook in VS Code
  - "CLT in action"



Consequence of the Central Limit Theorem:

The confidence interval for μ also applies to non-normal data:

Confidence intervals for the mean can be calculated based on the t-distribution in almost all situations, as long as n is "large enough." Why?: Assumptions for the t-test:(1) observations are independent (2) for small samples, the underlying population from which the sample is drawn should be approximately normally distributed. For larger samples, t-test applicable for CLT.

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When is n "large enough"?

Difficult to give a precise answer, BUT:

- Rule of thumb: $n \geq 30$
- Even for smaller n, the formula may be (almost) valid for non-normal data.

(Note: CLT applies to **means** – not variance)

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Example: Exam Question from 2016

Exercise IX

A course at a university is offered each semester typically with more than 300 students taking the exam. Examination results for 280 students who have passed the course at the previous exam is given in the table below. For example, the tables shows that 24 students got the grade 12. The distribution of the 280 grades is considered in the next 4 questions.

Grade	02	4	7	10	12	In total
Count	22	78	84	72	24	280

The data (grades) can be loaded into R by:

```
grades = rep(x=c(2,4,7,10,12), times=c(22,78,84,72,24))
```

Question IX.1 (12)

Use the central limit theorem to determine a 95% confidence interval for the mean grade based on the students who have passed the exam. (It is important in this question that the grades are perceived numerically, eg. 02 corresponds to the number 2, etc.).

Python: Exam Question from 2016

- Go to today's Python notebook in VS Code
  - "Example: Exam question from 2016"



Example: Tablet Production

Tablet Production:

In the production of tablets, an active ingredient is mixed with a powder, after which the mixture is formed into tablets. The goal is to produce a homogeneous mixture so that the strength of the tablets is consistent.

We consider a mixture of the active ingredient and filler powder, from which we aim to produce a large number of tablets.

We want the concentration of the active ingredient in the tablets to be 1 mg/g with the least possible variation. A random sample of 20 measurements is taken, each corresponding to what becomes one tablet. In each sample, the concentration of the active ingredient (in mg/g) is measured. Furthermore, we assume that our measurements follow a normal distribution.

Now, we want to estimate the variance of the concentration of the active ingredient and provide a confidence interval for this estimate.

From the sample of 20 measurements (representing 20 tablets), we find that the average concentration is 1.01 mg/g, with a sample standard deviation of 0.07 mg/g.

Python: Confidence Interval for Variance Estimate

Kahoot!

(x5)

- Go to today's Python notebook in VS Code
  - "Example: Production of tablets"



Agenda

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